RAFFLES INSTITUTION
2026 YEAR 6 TERM 2 TIMED PRACTICE

CANDIDATE NAME

CLASS 26

COMPUTING 9569

Practical **1 hour 15 minutes**

Additional materials: Electronic version of `USERS.txt` data file
Electronic version of `PHASEBOOK.txt` data file
Electronic version of `VIEWTUBE.txt` data file
Electronic version of `WHATSUP.txt` data file
Electronic version of `USERS.json` data file
Electronic version of Quick Reference Guide

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.

All tasks must be done in the computer laboratory. You are not allowed to bring in or take out any pieces of work or materials on paper or electronic media or in any other form.

Approved calculators are allowed.

Save each task as it is completed.

The use of built-in functions, where appropriate, is allowed for this paper unless stated otherwise.

Note that up to 4 marks out of 40 will be awarded for the use of common coding standards for programming style.

The number of marks is given in brackets [ ] at the end of each task.
The total number of marks for this paper is 40.

FOR EXAMINER'S USE		
Task 1	Task 2	TOTAL
		40

This document consists of **5** printed pages and **1** blank page.

Instruction: Start each task by labelling with a #. For example, #Task 1.1.

1 A tech company keeps a record of users on its social media and messaging platforms using a SQL database. The tech company offers three platforms for its users – Phasebook, Viewtube and Whatsup. When a user registers on one of its platforms, the following information is recorded.

Information	Description
UserID	Unique user ID assigned to user, in format Uddddddd, where d is a digit, for example, U123456
Name	Name of user
Email	Email address of user
Gender	Gender of user ('M' for male, 'F' for female)
DateOfBirth	Date of birth of user, in format dd/mm/yyyy
YearOfRegistration	Year that user first registered for an account
ActiveUser	Whether user is active or not (1 means user is active and 0 means user is not active)

For each platform, a "DateLogin" information will be recorded. This is the date that a user last logged in to the platform. Besides this, the follow extra information is recorded.

Platform	Information	Description
Phasebook	Friends	The number of friends of a user on Phasebook
Viewtube	Subscribers	The number of subscribers of a user on Viewtube
Whatsup	Contacts	The number of contacts of a user on Whatsup

The information is to be stored in four different tables:

- Users
- Phasebook
- Viewtube
- Whatsup

Task 1.1

Create an SQL file called TASK1_1_<your name>_<class>.sql to show the SQL code to create the database social.db with the four tables. The table, Users, must use UserID as its primary key. Each of the tables Phasebook, Viewtube and Whatsup should use UserID as both a primary key and foreign key referencing Users(UserID).

Save your SQL code as TASK1_1_<your name>_<class>.sql [5]

Task 1.2

The files `USERS.txt`, `PHASEBOOK.txt`, `VIEWTUBE.txt` and `WHATSUP.txt` contain information about the company's users for insertion into its database. Each row in the four files is a comma-separated list of information about the users.

For `USERS.txt`, information about each user is given in the following order:
`UserID,Name,Email,Gender,DateOfBirth,YearOfRegistration,ActiveUser`

For `PHASEBOOK.txt`, information about each user is given in the following order:
`UserID,Friends,DateLogin`

For `VIEWTUBE.txt`, information about each user is given in the following order:
`UserID,Subscribers,DateLogin`

For `WHATSUP.txt`, information about each user is given in the following order:
`UserID,Contacts,DateLogin`

Write a Python program to insert all information from the four files into `social.db` database. Run the program. Save your program as `TASK1_2_<your name>_<class>.py` [5]

Task 1.3

Write SQL code to show **only the `UserID`** of active users, sorted in ascending order.

Save your SQL code as `TASK1_3_<your name>_<class>.sql` [4]

Task 1.4

Write SQL code to show **only the names** of `Phasebook` users who registered after 2021, sorted in descending order.

Save your SQL code as `TASK1_4_<your name>_<class>.sql` [4]

2 After some time, the tech company decided to conduct a trial to keep a record of its users using a **NoSQL database**. The tech company offers three platforms for its users – Phasebook, Viewtube and Whatsup. A user may be registered on either one, two or three of the platforms. Information about its users is stored in the JSON file `USERS.json`. The following fields are recorded:

- Unique userID of user, in the format `Udddddd`, where `d` is a digit, for example, `U123456`
- Name of user
- Email address of user
- Gender of user (‘`M`’ for male, ‘`F`’ for female),
- Date of birth of user, in the format `dd/mm/yyyy`
- Year that user first registered an account
- Whether the user is active or not (1 means “active”, 0 means “not active”)
- Number of Phasebook friends
- Number of Viewtube subscribers
- Number of Whatsup contacts

Task 2.1

Using Jupyter Notebook platform, write a Python program to import the information from the JSON file into a MongoDB database. Save the information under the `users` collection in the `social` database. Ensure that only the fields from the JSON file are stored in the collection (no additional fields).

Save your program code as `TASK2_<your name>_<class>.ipynb` [4]

Task 2.2

Write program code that allows a data entry clerk to insert information of new users into the MongoDB database. The information to be inserted is the user’s userID, name, email address, gender, date of birth, year of registration (which is set to 2026 initially), and whether a user is active (which is set to 1 initially).

The other information about number of Phasebook friends, number of Viewtube subscribers and number of Whatsup contacts are not available, hence **should not** be inserted.

Use the following field names in the database: **`userID`**, **`name`**, **`email`**, **`gender`**, **`dateOfBirth`**, **`yearOfRegistration`**, **`active`**, **`phasebookFriends`**, **`viewtubeSubscribers`**, **`whatsupContacts`**.

Your program should use a loop to allow repeated input until the clerk chooses to quit. No need to validate input values.

Add your program code to `TASK2_<your name>_<class>.ipynb`

Run your program and insert these 2 documents:

"userID"	U862943	U269178
"name"	Mei Ling	Wei Chen
"email"	mei.ling39@yahoo.com	wei.chen44@yahoo.com
"gender"	F	M
"dateOfBirth"	06/01/2004	31/03/1999
"yearOfRegistration"	2026	2026
"active"	1	1

[6]

Task 2.3

Write program code that will allow the data entry clerk to choose to display users who have the corresponding field present (e.g. **`phasebookFriends`** for **`Phasebook`** users). The total number of documents should be displayed after the list of selected users.

Display the following menu for the clerk to choose options 1, 2, 3 or 4:

```
Display Users
[1] Phasebook Users
[2] Viewtube Users
[3] Whatsup Users
[4] Quit
```

The program should repeatedly display the menu until option 4 is selected.

If the data entry clerk chooses to display users of `Phasebook`, the output would look like this:

```
UserID     Name       Number of Phasebook friends
U755545    Jin Park   68
……         ……         ……
Number of documents = ……
```

When displaying users of `Viewtube`, the output would look like this:

```
UserID     Name       Number of Viewtube subscribers
U755545    Jin Park   12
……         ……         ……
Number of documents = ……
```

When displaying users of `Whatsup`, the output would look like this:

```
UserID     Name       Number of Whatsup contacts
U755545    Jin Park   238
……         ……         ……
Number of documents = ……
```

Add your program code to TASK2_<your_name>_<class>.ipynb

Run your program and select each of the 4 options in turn. [8]

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